Chapitre 02

Elements of probability:

I. Basic definitions:

Event: an outcome or Set of outcomes (حدث / مجموعة من النتائج)

Random experiment (التجربة العشوائية) is an experiment whose outcome can not be predicted with certainty, before the experiment is run. We usually assume (نفترض) that the experiment can be repeated infinitely under essentially the same conditions.

Random variables: the variables whose exact value can not be predicted.

Check Point
- dice = النرد
- Coin = عملة معدنية
- Throwing = رمي

Discrete (متقطع)
countable set of possible outcomes (مجموعة قابلة للعد من النتائج المحتملة)
ex: the number of children in a family

Continuous
unbroken chain of possible outcomes (سلسلة غير منقطعة من النتائج المحتملة)
ex: Leaf areas of the 4th leaf of a given plant (مساحات الأوراق للورقة الرابعة لنبات معين)

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Sample space (فضاء العينات): the set of all possible outcomes of a chance experiment it is denoted by Ω

II. Probability

Probability: is the likelihood or chance (احتمال) that a particular event will occur (يحدث)

P = number of possible points of the event / number of all possible point

$$P(A) = \frac{\text{عدد حالات الحادثة } A}{\text{مجموع الامكانيات (عدد الحالات الممكنة)}} \qquad P(\Omega) = 1$$

axioms of Probability (بديهيات)

- The probability is always between 0 and 1

Impossible event: an event that has no chance of occuring (Probability = 0)

Certain event: an event that is sure to occur (Probability = 1)

Equally likely outcomes: if each outcome (نتيجة) in an experiment has the same chance to occur

Mutually exclusive events: (حادثان متنافيتان: لا يقعان في نفس الوقت)

Events that cannot occur simultaneously

$$P(A \cup B) = P(A) + P(B)$$

Non mutually exclusive event (حادثان غير متنافيتان: يمكن حدوثهما في نفس الوقت)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

complement of an event A (العادث المكمل: العكس)

all events that are not part of event A

$$P(X) = 1 - P(A)$$

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I- Independent events: الأحداث المستقلة

Tow or more events are independent if the occurrence or non-occurrence of either of one does not affect the probability of the occurrence of the other
(يكون إذا كان حدوث أو عدم حدوث حادثة متصل لا يؤثر على احتمال حدوث الأخرى)

$$P(A \cap B) = P(A) \times P(B)$$

III- Laws of Probability:

Probability Distribution of a discret random variable التوزيع الاحتمالي للمتغير العشوائي المتقطع

is a list of probabilities associated (مرتبطة) with each of its possible values, it is also sometime called the probability function or the probability mass function (دالة الكتلة الاحتمالية)

$f(x)$ given by $f(x) = P(A \cap x)$, this is also denoted by $P(x = x)$

كل متغير نعطوه احتمال.

The table given below represent the probability of x

Comulative Probability الاحتمالات التراكمية

it is defined as the probability of observing less than or equal a given number of success
يتم تعريفه على أنه احتمال ملاحظة أقل من أو يساوي عدد معين من النجاح

Probability distribution of continuous random variables: Random variable x is continous if there is an integrable function (دالة قابلة للتكامل)

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- The function f_x is called the probability density function (pdf)

- Continuous random variable is not difined at specific variables, instead it is defined, it is defined over an interval of values, and is represented by the area under a curve

- لم يتم تعريف المتغير العشوائي عند قيم محددة بل تم تعريفه على مدى فترة من القيم ويتم تمثيلها بالمساحة الواقعة تحت المنحنى

Area under the curve

Areas = Probabilities

Top figure: histogram, ages ≤ 5 shaded

Bottom figure: pdf, ages ≤ 5 shaded

Both represent proportion of population ≤ 5

كلاهما يمثل نسبة من السكان

IV. Parameters of a distribution:

Expected Value (mean) - μ: the sum of the probability of each possible outcome of the experiment multiplied by the outcome value

Variance of a distribution ( ) - Var (x)

$Var(a) = 0$.

$Var(ax) = a^2 Var(x)$

$Var(x+y) = Var(x) + Var(y)$

σ^2

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Standard diviation:

is defined as the square root of variance, this indicator has more benefits than the variance in interpreting results.

Check Point
Sample mean = $\bar{x}$
Sample variance = S^2

V- Type of probability distributions

Descret probability distributions

Bernoulli distribution:

Describes a simple random experiment, with 2 possible outcomes success (1) نجاح
failure (0) فشل

- The probability of succes is the same for each trial احتمال النجاح ثابت في كل المحاولات
- The outcomes from different trails are indipendant

- There are a fixed number of trails conducted
n = عدد المحاولات في برنولي

$P(x=1) = p$

$P(x=0) = q = 1-p$

Mean Value

$E(x) = q0 + p.1 = p$

Variance $Var(X) = pq$ $\quad \sigma = \sqrt{pq}$

Remarque

The experiment of Bernoulli carried out n times

Binomial distribution

the experiment of Bernoulli carried out n times

mean = np Variance = npq Standard deviation $= \sqrt{npq}$

Symbols

n: number of times a trial is repeated

$p = P(s)$ احتمال حدوث النجاح

$q = P(F)$ احتمال عدم حدوث الفشل

$q = 1 - p$

X: The random variable Succes عدد مرات النجاح في n trials

$P(X = k)$

k: number of succes desired

For proportion (%)

$$\text{Mean} = \mu = \frac{nP}{n} = P$$

$$\text{Var}(\%) = \frac{P(1-P)}{n}$$

Poisson distribution

The binomial distribution with n very large and at the same time p very small

Remarque :

- عندما يكون عدد التجارب n من المتوسط نأخذ الكبير $n > 30$ يمكن تقريب التوزيع ذي الحدين normal distribution. بشكل

at most : على الأكثر

أقل أو يساوي الناتج.

at least : على الأقل

أكبر أو يساوي الناتج.

عندما نطلب Cumulative من الجدول لازم يكون at most لكن إذا كانت at least نحسب P(s) ونطرح $1 - P(s)$.



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